

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase III, Randomized, Double-blind Study to Evaluate Pembrolizumab Plus Chemotherapy vs Placebo Plus Chemotherapy as Neoadjuvant Therapy and Pembrolizumab vs Placebo as Adjuvant Therapy for Triple Negative Breast Cancer (TNBC)
Short Title:	Study of Pembrolizumab (MK-3475) Plus Chemotherapy vs Placebo Plus Chemotherapy as Neoadjuvant Therapy and Pembrolizumab vs Placebo as Adjuvant Therapy in Participants With Triple Negative Breast Cancer (TNBC) (MK-3475-522/KEYNOTE-522)
Protocol Number:	851420532061182204
NCT Number:	NCT03036488
Sponsor Name:	Merck
Investigational Product:	cyclophosphamide
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Pathological complete response (pCR) rate using the definition of ypT0/Tis ypN0 (i.e., no invasive residual in breast or nodes; noninvasive breast residuals allowed) at the time of definitive surgery: pCR rate (ypT0/Tis ypN0) is defined as the percentage of participants without residual invasive cancer on hematoxylin and eosin evaluation of the complete resected breast specimen and all sampled regional lymph nodes following completion of neoadjuvant systemic therapy by current American Joint Committee on Cancer (AJCC) staging criteria assessed by the local pathologist at the time of definitive surgery. 2. Event-free Survival (EFS) as assessed by Investigator: EFS is defined as the time from randomization to any of the following events: progression of disease that precludes surgery, local or distant recurrence, second primary malignancy (breast or other cancers) or death due to any cause.
Secondary Obj.:	1. pCR rate using an alternative definition, ypT0 ypN0 (i.e., no invasive or noninvasive residual in breast or nodes) at the time of definitive surgery 2. pCR rate using the definition of ypT0/Tis ypN0 (i.e., no invasive residual in breast or nodes; noninvasive breast residuals allowed) at the time of definitive surgery 3. EFS in participants with tumors expressing PD-L1

2.2 Background & Rationale

The purpose of this study is to evaluate the efficacy and safety of pembrolizumab (MK-3475) plus chemotherapy vs placebo plus chemotherapy as neoadjuvant therapy and pembrolizumab vs placebo as adjuvant therapy in participants

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who have triple negative breast cancer (TNBC).

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: COMPLETED

Target N: 1174

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Has newly diagnosed, locally advanced, centrally confirmed TNBC, as defined by the most recent American Society of Clinical Oncology (ASCO)/College of American Pathologists (CAP) guidelines.
- * Has previously untreated locally advanced non-metastatic (M0) TNBC defined as the following combined primary tumor (T) and regional lymph node (N) staging per current American Joint Committee of Cancer (AJCC) staging criteria for breast cancer as assessed by the investigator based on radiological and/or clinical assessment:

- * T1c, N1-N2
- * T2, N0-N2
- * T3, N0-N2
- * T4a-d, N0-N2

- * Provides a core needle biopsy consisting of at least 2 separate tumor cores from the primary tumor at screening to the central laboratory.

- * Has Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 performed within 10 days of treatment initiation.

- * Demonstrates adequate organ function.

- * Males and female participants of childbearing potential must be willing to use an adequate method of contraception for the course of the study through 12 months after the last dose of study treatment for participants who have received cyclophosphamide, and 6 months after the last dose of study treatment for participants who did not.

4.2 Exclusion Criteria

- * Has a history of invasive malignancy ≥5 years prior to signing informed consent except for adequately treated basal cell or squamous cell skin cancer or in situ cervical cancer.

- * Has received prior chemotherapy, targeted therapy, and radiation therapy within the past 12 months.

- * Has received prior therapy with an anti-programmed cell death protein 1 (anti-PD-1), anti-programmed death - ligand 1 (anti-PD-L1), or anti-PD-L2 agent or with an agent directed to another co-inhibitory T-cell receptor (e.g., cytotoxic T-lymphocyte-associated antigen-4 [CTLA-4], OX-40, CD137 [tumor necrosis factor receptor superfamily member 9 (TNFRSF9)]) or has previously participated in a pembrolizumab (MK-3475) clinical study.

- * Is currently participating in or has participated in an interventional clinical study with an investigational compound or device within 4 weeks of the first dose of treatment in this current study.

- * Has received a live vaccine within 30 days of the first dose of study treatment.

- * Has an active autoimmune disease that has required systemic treatment in past 2 years (i.e., with use of disease modifying agents, corticosteroids or immunosuppressive drugs).

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- * Has a diagnosis of immunodeficiency or is receiving systemic steroid therapy (i.e., dosing exceeding 10 mg daily of prednisone or equivalent) or any other form of immunosuppressive therapy within 7 days prior to the first dose of study treatment.
- * Has a known history of Human Immunodeficiency Virus (HIV).
- * Has known active Hepatitis B or Hepatitis C.
- * Has a history of (non-infectious) pneumonitis that required steroids or current pneumonitis.
- * Has an active infection requiring systemic therapy.
- * Has significant cardiovascular disease, such as: history of myocardial infarction, acute coronary syndrome or coronary angioplasty/stenting/bypass grafting within the last 6 months OR congestive heart failure (CHF) New York Heart Association (NYHA) Class II-IV or history of CHF NYHA Class III or IV.
- * Is pregnant or breastfeeding, or expecting to conceive children within the projected duration of the study, starting with the screening visit through 12 months after the last dose of study treatment for participants who have received cyclophosphamide, and for 6 months after the last dose of study treatment for participants who have not.
- * Has a known hypersensitivity to the components of the study treatment or its analogs.
- * Has a known history of active tuberculosis (TB, Bacillus Tuberculosis).

11. Study Identifiers & Governance

Primary ID: 851420532061182204

Registry Number: NCT03036488

Sponsor: Merck

Study Title: Study of Pembrolizumab (MK-3475) Plus Chemotherapy vs Placebo Plus Chemotherapy as Neoadjuvant Therapy and Pembrolizumab vs Placebo as Adjuvant Therapy in Participants With Triple Negative Breast Cancer (TNBC) (MK-3475-522/KEYNOTE-522)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____